

PRADEEP PONNAM

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EDUCATION

The University of Memphis

Master of Science, Information Technology

Aug 2024 - May 2026

- **GPA:** 4/4 (A+)
- **Achievements:** Engineering GPA: 4/4 (A+)
- **Coursework:** Machine Learning, Generative AI, Information Systems Security, Software Engineering

WORK EXPERIENCE

ISRO

Jan 2024 - Apr 2024

Research Intern

- Engineered a deep learning prototype using neural networks to detect ships and oil spills from SAR satellite imagery, emphasizing clean, modular Python code and GPU acceleration.
- Created a modular training pipeline in Python with live visualization using TensorBoard, utilizing version control best practices.
- Collaborated on an innovation challenge within the Earth Observation division, contributing to a proposal for integration with coastal surveillance platforms alongside DRDO.
- Iterated on the system design based on team feedback, ensuring maintainability and scalability of AI components.

University of Memphis

Apr 2025 - Present

Research Assistant

- Developed a GPT-4 chatbot using FastAPI, expertise in applying LLM APIs to assist executives in navigating US AI regulations.
- Integrated BERT embeddings to match legislation with user profiles geography, demonstrating robust machine learning knowledge.
- Designed smart onboarding flows and prompt-tuned response logic for regulatory training, ensuring clear and efficient software.
- Piloted the chatbot with academic and enterprise users, iterating features based on proactive feedback and agile methodologies.

Slash Mark

Feb 2024 - May 2024

Java Backend Intern

- Developed REST APIs using Spring Boot + MySQL
- Implemented secure JWT-based authentication & role-based access
- Deployed Dockerized microservices on Heroku with CI/CD (GitHub Actions)

BlackBuck

May 2024 - Aug 2024

Software Engineer Intern

- Engineered a scalable backend microservice architecture using Spring Boot and Kafka to handle event
- Developed internal APIs to support logistics coordination for 1,000+ users across web and mobile platforms
- Implemented automated deployment workflows using Docker, GitHub Actions, and AWS EC2

PERSONAL PROJECTS

AI Regulation Chatbot.

- GPT-4 chatbot using FastAPI & Firebase to guide executives on AI laws across U.S. states.
- Integrated BERT + FAISS for legal text matching and Gemini for summarization.

Enterprise Document Processor.

- OCR-based platform built with Spring Boot, Kafka, AWS S3, and BERT for extracting entities from scanned PDFs.
- Used JWT-secured APIs and Dockerized microservices.

Drone Vision Object Detection.

- Aerial object detection system using GLSAN + YOLOv5 to identify vehicles and infrastructure in drone footage.
- Benchmarked against CNNs with 20% mAP gain.

Java HR Microservice.

- Spring Boot + MySQL REST API for HR operations with JWT login, Swagger UI, and CI/CD via GitHub Actions.

SKILLS

- **Programming Languages:** Java (Object Oriented Programming), Python (AI, ML, DS), C, R, JavaScript, HTML/CSS, SQL
- **Tools & Technologies:** GIT, PostgreSQL, LLM APIs, Application Deployment, Cloud Services, Education Technology
- **Methodologies & Communication:** Agile Methodology, Data Structures, Algorithms, English Language Skills

AWARDS

- **Top 1% Performer ISRO Research Cohort (2024):** Recognized for developing a YOLOv5-based oil spill detection system used in SAR imagery analysis.
- **Dean's List University of Memphis (2024, 2025):** For consistent academic excellence with a 4.0 GPA in the MS MIS program.